RAJALAKSHMI ENGINEERING COLLEGE
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INSTITUTION'S INNOVATION COUNCIL
(Ministry of HRD Initiative)

DEPARTMENT OF CSE

AI-Powered Team Formation Tool

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ABSTRACT

The **AI-Powered Team Formation Tool** is an intelligent application designed to automatically create fair, balanced, and skill-diverse teams based on individual skills, attendance, portfolio submission, and leadership experience. It uses **Artificial Intelligence techniques such as a Forward-Chaining Rule Engine and Constraint Satisfaction Problem (CSP) solving** to determine role eligibility and generate suitable teams.

The system allows users to specify team sizes, required skills, eligible roles, and team member constraints. A backtracking algorithm finds team combinations that satisfy these requirements while maintaining fairness and skill diversity. It also provides fairness metrics and identifies unmet constraints when a perfect team arrangement is not possible. Additionally, the system suggests suitable campus meeting points for the formed teams.

The application is developed using **Python and Streamlit**, providing a simple and interactive user interface. This tool helps reduce manual effort, improve team organization, and support efficient collaboration in academic and project-based environments.

PROBLEM STATEMENT

Forming balanced and effective teams manually is a challenging task, as it is difficult to evaluate individual skills, attendance, portfolio contributions, and leadership experience fairly. Traditional team formation methods may result in uneven skill distribution, unbalanced workloads, and unsuitable role assignments.

Therefore, an AI-powered team formation tool is needed to automatically organize individuals into balanced teams based on predefined requirements and constraints. The system aims to improve fairness, maximize skill diversity, assign suitable roles, and reduce the time and effort required for team formation.

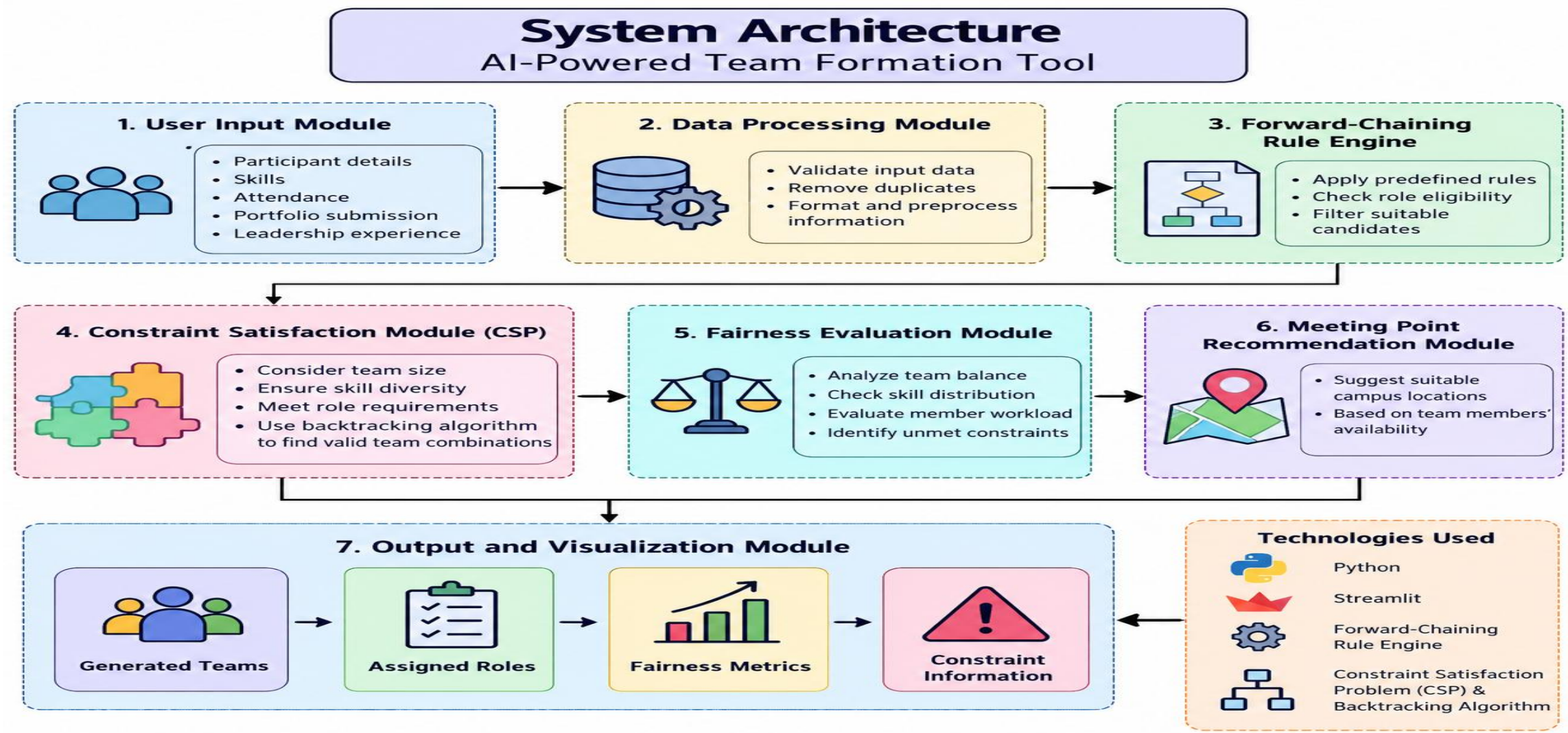
LITERATURE SURVEY

S. No.	Author(s) & Year	Research Paper Title	Methodology	Limitation
1	J. Wang et al. (2016)	USTF: A Unified System of Team Formation	Compares team formation algorithms and evaluates their performance using real-world datasets.	Focuses mainly on team formation in social networks and algorithm benchmarking.
2	Robert Bredereck et al. (2018)	Parameterized Complexity of Team Formation in Social Networks	Studies the computational complexity of forming skill-based teams while minimizing communication costs.	Finding suitable teams can become computationally challenging as constraints increase.

LITERATURE SURVEY

S. No.	Author(s) & Year	Research Paper Title	Methodology	Limitation
3	Kalyani Selvarajah et al. (2021)	A Unified Framework for Effective Team Formation in Social Networks	Uses a multi-objective approach considering expertise, communication costs, trust, and proximity.	The framework involves multiple objectives and may require complex optimization.
4	Diego Gómez-Zará et al. (2022)	In Search of Diverse and Connected Teams: A Computational Approach to Assemble Diverse Teams Based on Members' Social Network	Uses NSGA-II, a multi-objective genetic algorithm, to form diverse teams while considering existing social connections.	Requires additional information about members' characteristics and social connections.

ARCHITECTURE FLOW DIAGRAM



PROPOSED SYSTEM

The proposed system automatically forms balanced and effective teams using an AI-based, rule-driven approach. It considers participants' skills, attendance, portfolio submission, and leadership experience to create suitable teams.

The system uses a **Forward-Chaining Rule Engine** to identify eligible participants and a **Constraint Satisfaction Problem (CSP) with backtracking** to generate teams that satisfy predefined conditions. A fairness evaluation module checks skill distribution, role allocation, and team balance. It also recommends suitable campus meeting points for the formed teams.

IMPLEMETATION DETAILS

1. Frontend

Technology: Streamlit

Provides a user-friendly interface to enter participant details and view generated teams.

Displays team members, assigned roles, and fairness metrics.

2. Backend

Programming Language: Python

Processes participant data and manages team formation.

Implements rule-based decision-making and team allocation logic.

3. Database

Technology: SQLite

Stores participant details, skills, attendance, portfolio status, leadership experience, and generated team information.

IMPLEMETATION DETAILS

4. AI and Algorithms

Forward-Chaining Rule Engine: Identifies eligible participants and suitable roles based on predefined rules.

Constraint Satisfaction Problem (CSP): Ensures teams meet specified requirements.

Backtracking Algorithm: Searches for valid team combinations when constraints are not satisfied.

Key Features:

- Automatic team formation.
- Balanced distribution of skills among teams.
- Role assignment based on participants' abilities.
- Fairness evaluation and constraint checking.
- Identification of unmet constraints.
- Campus meeting-point recommendations.

RESULTS

The AI-Powered Team Formation Tool aims to automate the process of forming balanced and effective teams based on participants' skills, attendance, portfolio submission, and leadership experience. The system uses rule-based techniques and constraint satisfaction algorithms to distribute members, assign suitable roles, and evaluate team fairness. It also identifies unmet constraints and recommends suitable campus meeting points. The proposed system helps reduce manual effort, improve skill distribution, and support fair and efficient team formation through a user-friendly interface.

Outcome

The proposed system aims to reduce manual effort, improve team balance, and support fair and efficient team formation.

AI Team Formation Tool

Enter a list of people with their skills, set your constraints, and let the solver form fair, balanced teams automatically.

1. Enter People & Skills

Edit the table below — add/remove rows, and list skills comma-separated.

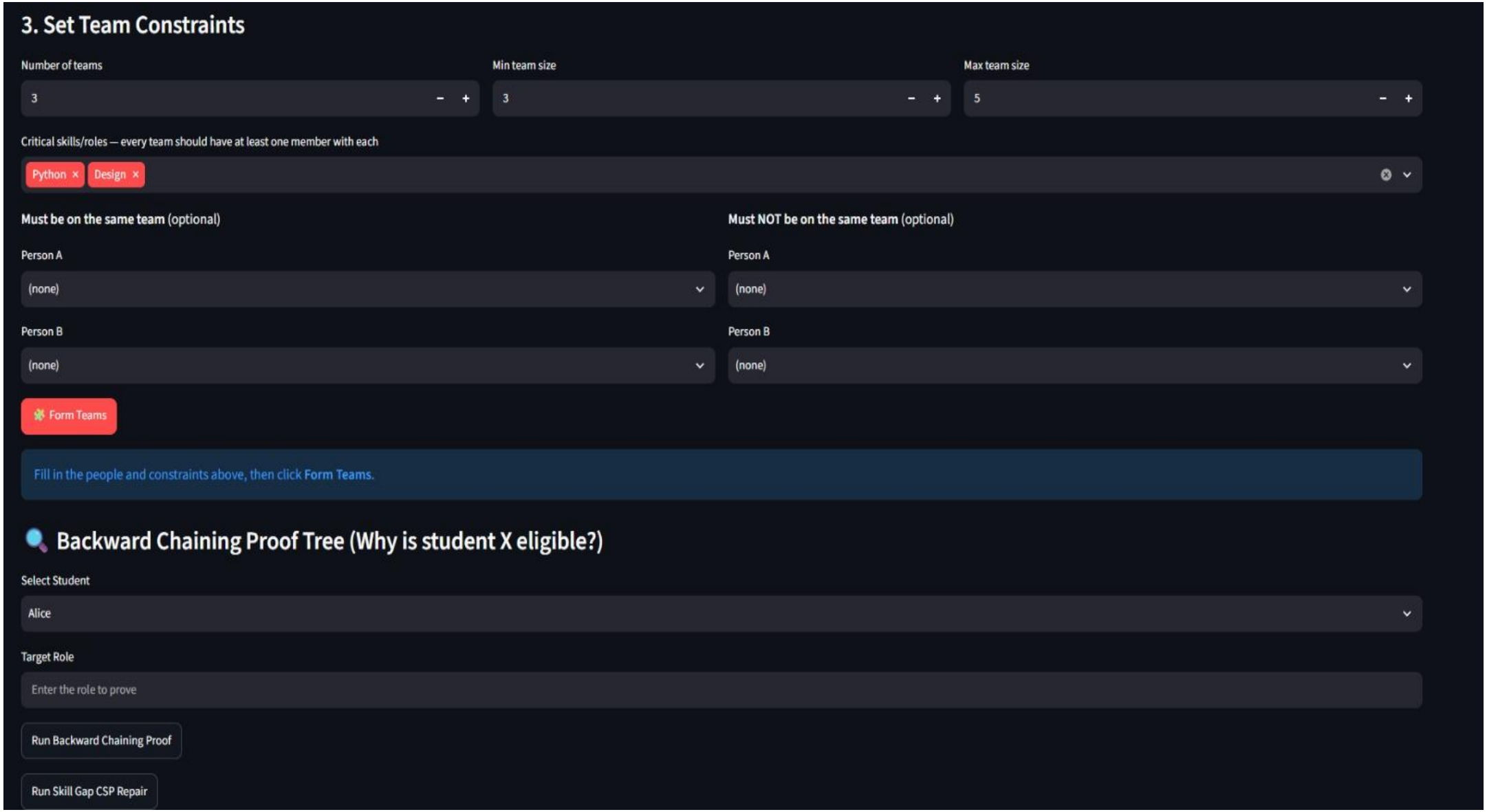
Name	Skills	Attendance %	Portfolio Submitted	Leadership Experience	Location
Alice	Python, ML	92	☐	☑	Hostel A
Bob	Design	70	☑	☐	Hostel B
Carol	Python, Writing	88	☐	☐	Library
Dave	Design, PM	95	☑	☑	Admin Block
Eve	Writing	65	☐	☐	Cafeteria
Frank	ML, PM	80	☐	☐	Sports Complex
Grace	Python	55	☐	☐	Lab Block
Heidi	Design, Writing	91	☑	☐	Auditorium
Ivan	Testing, DevOps	78	☐	☐	Parking Lot
Judy	Python, Testing	89	☐	☐	Food Court

2. Eligibility Check (Logic Engine)

A forward-chaining rule engine derives which roles each person qualifies for, based on their skills, attendance, portfolio, and leadership experience.

Name	Eligible Roles
Alice	Developer, TeamLead
Bob	Designer
Carol	Developer, Writer
Dave	Designer, ProjectManager, TeamLead
Eve	Writer
Frank	–
Grace	–
Heidi	Designer, Writer
Ivan	–
Judy	Developer

› View reasoning trace for a person


3. Set Team Constraints
Number of teams
3
Min team size
3
Max team size
5
Critical skills/roles — every team should have at least one member with each
Python
Design
Must be on the same team (optional)
Person A
(none)
Person B
(none)
Must NOT be on the same team (optional)
Person A
(none)
Person B
(none)
Form Teams
Fill in the people and constraints above, then click Form Teams.
Backward Chaining Proof Tree (Why is student X eligible?)
Select Student
Alice
Target Role
Enter the role to prove
Run Backward Chaining Proof
Run Skill Gap CSP Repair

FUTURE ENHANCEMENT

Machine Learning Integration: Predict team performance based on members' skills and previous project experiences.

Advanced Skill Assessment: Evaluate participants' technical and soft skills for better team formation.

Real-Time Availability: Track members' availability to form suitable teams.

Automatic Role Assignment: Assign roles based on individual strengths and expertise.

Meeting Schedule Recommendation: Suggest suitable meeting times and campus locations.

Cloud-Based Deployment: Make the system accessible from anywhere.

Mobile Application: Develop a mobile app for easier access.

College Management Integration: Connect with college systems to collect participant information automatically.

REFERENCE

Wang, J. et al. (2016), *USTF: A Unified System of Team Formation*, IEEE. https://ieeexplore.ieee.org/document/7445184

Bredereck, R. et al. (2018), *Parameterized Complexity of Team Formation in Social Networks*, Theoretical Computer Science. https://doi.org/10.1016/j.tcs.2017.05.024

Selvarajah, K. et al. (2021), *A Unified Framework for Effective Team Formation in Social Networks*, Expert Systems with Applications. https://doi.org/10.1016/j.eswa.2021.114886

Gómez-Zará, D. et al. (2022), *In Search of Diverse and Connected Teams: A Computational Approach to Assemble Diverse Teams Based on Members' Social Networks*, PLOS ONE. https://doi.org/10.1371/journal.pone.0276061

Python Documentation

Streamlit Documentation

SQLite Documentation

THANK YOU